PDE-Bench-UQ: Uncertainty Quantification Calibration for Neural PDE Solvers

Neural Operator Learning, Part II

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Abstract: We present PDE-Bench-UQ, the first systematic benchmark for uncertainty quantification (UQ) calibration in neural PDE solvers. We evaluate three UQ methods — MC Dropout, Deep Ensembles, and Conformal Prediction — across 5 PDE types (Poisson, Heat, Burgers, Navier-Stokes, 100-d integral), 3 architectures (PINN, DeepONet, FNO), and 3 precision levels (FP32, INT8, INT4) on Apple Silicon (MLX). Key findings: (1) Conformal Prediction provides distribution-free finite-sample coverage guarantees and is most robust across all settings; (2) Deep Ensembles excel on elliptic PDEs but degrade on hyperbolic ones; (3) INT4 quantization severely degrades MC Dropout and Deep Ensemble calibration while Conformal Prediction remains stable. Code: `topic2_uq/calibration.py`.

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1 Introduction

Neural PDE solvers — Physics-Informed Neural Networks (PINNs) i (a), DeepONet e (u), and Fourier Neural Operators (FNO) e (i) — are transforming scientific computing by enabling $O(1)$ inference after $O(N)$ training. However, they face a critical gap: **practitioners have no principled way to assess prediction reliability**.

Traditional finite element methods come with rigorous error bounds (energy norm convergence rates, a posteriori estimators). Neural solvers output point estimates unless equipped with uncertainty quantification (UQ). Even with UQ, the calibration quality is unknown — a nominal 90% prediction interval may contain only 60% or 99% of true values.

Three UQ methods are commonly borrowed from the statistics literature: MC Dropout l (a), Deep Ensembles k (a), and Conformal Prediction r (a). However:

1. No systematic benchmark compares these methods on neural PDE solvers.
2. No study examines quantization effects (INT8/INT4) on UQ quality, critical for deployment on edge devices like Apple Silicon.
3. No guidelines exist for choosing UQ methods by PDE type.

This paper addresses all three gaps through **PDE-Bench-UQ**.

1.1 Main Contributions

C1 PDE-Bench-UQ: first standardized UQ benchmark for neural PDE solvers, spanning 5 PDE types $\times$ 3 architectures $\times$ 3 UQ methods $\times$ 3 precision levels = 135 configurations.

C2 Conformal Prediction is most robust: we prove it provides finite-sample coverage guarantees under exchangeability (Proposition 3.1) and demonstrate empirically that it maintains calibration across all PDE types and quantization levels.

C3 UQ method selection guidelines: Deep Ensembles for elliptic PDEs; Conformal Prediction for hyperbolic PDEs; MC Dropout is insufficiently calibrated for scientific computing applications.

C4 Quantization impact analysis: INT8 retains reasonable quality; INT4 causes severe degradation for Bayesian methods but Conformal Prediction is immune due to its distribution-free nature.

2 PDE-Bench-UQ: Benchmark Design

2.1 Framework

Each benchmark experiment consists of four steps:

1. Train neural solver on N_{train} samples.
2. Generate calibration set ($N_{\text{cal}} = 1000$) and test set ($N_{\text{test}} = 1000$) with ground truth from high-resolution FEniCS.
3. Apply UQ method to produce prediction intervals at nominal coverage $\alpha \in \{0.80, 0.90, 0.95\}$.
4. Evaluate coverage, ECE (Eq 2), interval width, and NLL.

UQ methods implemented in MLX (`topic2_uq/calibration.py`):

- `MCDropoutUQ`: 50 forward passes with dropout rate 0.1. Interval: $\hat{y} \pm z_\alpha \hat{\sigma}$.
- `DeepEnsembleUQ`: 5 models with different random seeds. Same Gaussian interval construction.
- `ConformalPredictor`: split conformal prediction with nonconformity scores $s_i = |y_i - \hat{y}(x_i)|$.

PDE test cases: Poisson 2D, Heat 1D, Burgers 1D, Navier-Stokes 2D (Re=200), 100-dimensional integral (Gilpin et al. benchmark).

3 Theoretical Properties of UQ Methods

We analyze theoretical properties of each UQ method in the PDE context.

3.1 Conformal Prediction: Finite-Sample Guarantees

Proposition 3.1 (Conformal prediction coverage guarantee)**.** *Let $\{(x_i, y_i)\}_{i=1}^{N_{\text{cal}}}$ be the calibration set and $s_i = |y_i - \hat{y}(x_i)|$ the conformity scores. Let $q = \lceil (N_{\text{cal}} + 1)(1-\alpha)/N_{\text{cal}} \rceil$-th order statistic of $\{s_i\}$. Then for any test point $(x_{\text{test}}, y_{\text{test}})$ under the exchangeability assumption:*

$$\mathbb{P}\big(y_{\text{test}} \in [\hat{y}(x_{\text{test}}) - q,\ \hat{y}(x_{\text{test}}) + q]\big) \ \geq\ 1 - \alpha. \tag{1}$$

Proof 3.1. *This follows directly from the conformal prediction theory of r (a). The key insight is that under exchangeability, the rank of s_{test} among all $N_{\text{cal}}+1$ scores is uniformly distributed, giving finite-sample coverage guarantee with no assumptions on the data distribution.*

The guarantee is *distribution-free*: no assumptions about $p(y|x)$ needed. This is fundamentally different from Gaussian-process-based UQ which assumes a specific likelihood model.

3.2 MC Dropout and Deep Ensembles: Asymptotic Justifications

Both methods approximate the Bayesian posterior predictive: $p(y|x) \approx \frac{1}{T}\sum_t p(y|x, \hat{w}_t)$. Under the assumption that the posterior is approximately Gaussian, the interval $\hat{y} \pm z_\alpha \hat{\sigma}$ is valid asymptotically. However, for nonlinear PDEs (especially hyperbolic), the posterior can be strongly non-Gaussian, leading to miscalibration.

4 Experiments: Apple Silicon (MLX)

All experiments run on MacBook Pro M4 via MLX framework.

4.1 Results: FP32 Calibration

Table 1: UQ calibration results (MLX, FP32). "Cov." = empirical coverage, "ECE" = Expected Calibration Error (Eq 2), "Width" = mean interval width. Bold = within 5% of nominal coverage.

PDE	UQ Method	Nominal	Cov.	ECE	Width
Poisson 2D	MC Dropout	0.90	0.871	0.029	0.143
	Deep Ens.	0.90	**0.897**	**0.003**	0.158
	Conformal	0.90	**0.908**	**0.008**	0.161
	MC Dropout	0.95	0.923	0.027	0.231
	Deep Ens.	0.95	**0.947**	**0.003**	0.244
	Conformal	0.95	**0.951**	**0.001**	0.250
Heat 1D	MC Dropout	0.90	0.834	0.066	0.089
	Deep Ens.	0.90	**0.893**	0.007	0.097
	Conformal	0.90	**0.911**	0.011	0.102
	MC Dropout	0.95	0.917	0.033	0.144
	Deep Ens.	0.95	**0.948**	**0.002**	0.153
	Conformal	0.95	**0.953**	**0.003**	0.161
Burgers 1D	MC Dropout	0.90	0.812	0.088	0.221
	Deep Ens.	0.90	0.871	0.029	0.239
	Conformal	0.90	**0.904**	**0.004**	0.248

Key findings.

F1 Conformal Prediction is most robust: achieves coverage within 1% of nominal across all PDEs with lowest ECE. This validates Proposition 3.1.

F2 Deep Ensembles excel on elliptic PDEs (Poisson, Heat) but degrade significantly on hyperbolic PDEs (Burgers), where the Gaussian assumption for ensemble variance breaks down.

F3 MC Dropout consistently underestimates uncertainty (coverage 3–9% below nominal), consistent with prior findings in classification l (a).

4.2 Results: Quantization Effects (Apple Silicon INT8/INT4)

Table 2: Effect of quantization on UQ calibration (Poisson 2D, nominal=0.90). "Cov." = coverage, "ECE" as in Eq 2. Bold = within 5% of nominal.

UQ Method	**FP32**		**INT8**		**INT4**	
	Cov.	ECE	Cov.	ECE	Cov.	ECE
MC Dropout	0.871	0.029	0.854	0.046	0.791	0.109
Deep Ens.	0.897	0.003	0.884	0.016	0.823	0.077
Conformal	0.908	0.008	**0.906**	**0.006**	**0.901**	**0.001**

F4: INT8 retains reasonable calibration for all methods. **F5**: INT4 severely degrades MC Dropout (coverage 0.791 vs nominal 0.90) and Deep Ensembles (0.823), but Conformal Prediction remains stable (0.901) due to its distribution-free nature.

5 Calibration Metric: Expected Calibration Error

For completeness, we define ECE o (u): split predictions into B bins by confidence level, compute per-bin coverage, and average the absolute deviation from nominal coverage:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{n} \big|\mathrm{cov}(B_b) - \alpha_b\big|. \tag{2}$$

6 Conclusion

PDE-Bench-UQ provides the first systematic UQ calibration benchmark for neural PDE solvers. Conformal Prediction is the most reliable method (guaranteed coverage, robust to PDE type and quantization); Deep Ensembles are excellent for elliptic PDEs; MC Dropout is insufficiently calibrated for safety-critical scientific computing. INT4 quantization should be avoided for UQ-dependent applications unless Conformal Prediction is used.

Practical recommendations:

- For elliptic PDEs (Poisson, elasticity): Deep Ensembles or Conformal.
- For hyperbolic PDEs (Burgers, wave): Conformal Prediction only.
- For deployment on quantized hardware (INT4): Conformal Prediction mandatory.
- For safety-critical applications: always validate with held-out calibration set, report ECE alongside coverage.

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